

CO5 AT3 HACKATHON MINI PROJECT

SMART CURRICULUM ACTIVITY AND ATTENDANCE APP – DETAILED PROJECT REPORT WITH CODE

Login Module

Frontend (login.html):

```
<form method="post">
<input type="email" name="email" required>
<input type="password" name="password" required>
<select name="role">
<option value="student">Student</option>
<option value="teacher">Teacher</option>
</select>
<button>Login</button>
</form>
```

Backend (Flask):

```
@app.route('/login', methods=['GET','POST'])
def login():
    if request.method == 'POST':
        session['user'] = request.form['email']
        return redirect(url_for('dashboard'))
    return render_template('login.html')
```

Signup Module

Frontend (signup.html):

```
<form method="post">
<input name="name" required>
<input type="email" name="email" required>
```

```
<input type="password" name="password" required>
<button>Create Account</button>
</form>
```

Backend:

```
@app.route('/signup', methods=['GET','POST'])
def signup():
    if request.method == 'POST':
        return redirect(url_for('login'))
    return render_template('signup.html')
```

Dashboard Module

```
@app.route('/dashboard')
def dashboard():
    return render_template(
        'dashboard.html',
        subjects=subjects
    )
```

Subjects Module

```
@app.route('/subjects')
def subjects_page():
    return render_template(
        'subjects.html',
        subjects=subjects
    )
```

Quiz Module

```
@app.route('/quiz')
def quiz():
    return render_template('quiz.html')

question_banks = {
    'python-programming':[
        ('Which keyword defines a function in Python?',
         ['def','func','function'],
         'def')
    ]
}
```

Attendance Module

```
@app.route('/attendance')
def attendance():
    return render_template('attendance.html')
```

Attendance page displays:

- Present days
- Absent days
- Attendance percentage
- Attendance streak

Progress Tracking Module

```
@app.route('/progress')
def progress():
    return render_template('progress.html')
```

Tracks:

- Subject progress
- XP points
- Levels
- Completion percentage

Leaderboard Module

```
@app.route('/leaderboard')  
def leaderboard():  
    return render_template('leaderboard.html')
```

Displays rankings based on XP points.

Badges Module

```
@app.route('/badges')  
def badges():  
    return render_template('badges.html')
```

Achievement badges are awarded based on learning milestones.

Teacher Dashboard

```
@app.route('/teacher/dashboard')  
def teacher_dashboard():  
    return render_template('teacher/dashboard.html')
```

Features:

- Student monitoring
- Report generation

- Quiz management

Conclusion

The application integrates authentication, learning modules, quizzes, attendance monitoring, gamification, progress tracking, and teacher reporting into a single educational platform.